MARITIME (MRT)

Market-First → Maritime Settlement & Dispute Rail Roadmap, Workflows, and Risk Controls

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Core message
This roadmap is designed to fit real shipping practice:
- We do not rewrite Charter Parties (CP) or replace arbitration.
- We do not force full-hire escrow early.
- We standardize evidence, neutralize disputed portions, and execute agreed/binding outcomes.

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1. What exists today and why Market-First

Shipping settlements still rely on emails, Excel spreadsheets, and non-standard evidence packs. Disputes often become a cashflow and leverage problem: whoever holds the money has bargaining power. Market-First establishes credibility (transparent token market, documentation, and wallet proofs) before moving into utility modules.

Value proposition
MRT's advantage is not 'replacing banks'. It reduces negotiation friction by:
1) creating a shared timeline (Deal Ledger),
2) standardizing evidence (Evidence Standardization Protocol), and
3) neutralizing disputed portions via narrow-purpose vaults (no leverage).

Market-first outputs (committed)

- Beginner guides (EN/TR): verify contract, buy/sell safely, avoid counterfeit tokens.
- Transparency artifacts: official wallets, allocations, reward budgets, on-chain proofs.
- Liquidity incentives that reward liquidity + time (not trading volume) to avoid wash-trading.

2. What MRT builds first (MVP scope)

The MVP is shipping-practice compatible. Hire and freight can continue through existing bank rails. MRT adds a neutral settlement layer on top.

MVP building blocks

- Deal Ledger: one shared deal page per fixture/CP with a complete timeline and audit trail.
- Evidence Vault (off-chain): structured upload of required documents; optional hash anchoring.
- Disputed Portion Vaults: narrow-purpose USDC vaults used only for the disputed amount (not full hire).
- Settlement Workflow: proposals, counteroffers, negotiation windows, and escalation to arbitration.

Key shipping-compatible rule
MVP default: 'deduction → vault'. If a party would normally deduct or withhold, MRT asks them to place the disputed portion into a neutral vault instead of keeping it in their own account.

3. Trust & governance model

MRT does not decide who is right. The vault is a custody and execution layer. Legal interpretation remains with the CP and the chosen resolution process. The design prevents unilateral seizure or withholding of disputed funds.

3.1 Signers and approvals (2-of-3)

Default releases require 2-of-3 approvals. Typical signers: Owner, Charterer, and a neutral role. In the MVP, the neutral role is NOT a judge; it acts as a Decision Executor and procedure checker.

- Owner + Charterer: fastest path when both agree.
- Party + Decision Executor: executes a binding decision (arbitration/court) or confirms procedure.
- Decision Executor cannot move funds alone.

3.2 Negotiation and escalation timelines

Stage	Typical window	Outcome
Owner response window	3–5 days	Accept / Reject / Partial + upload counter-evidence
Negotiation window	7–14 days	Proposals & counteroffers recorded in Deal Ledger
Escalation trigger	After window	Case file export; proceed via CP venue (e.g., arbitration)
Execution	On decision	Party + Decision Executor executes settlement from vault

3.3 Why this is fair (no side advantage)

Traditional workflows often reward the side holding cash. MRT removes that advantage by using neutral custody for the disputed portion. Either side can initiate a vault line item, and unused amounts are returned to the original depositor(s) pro-rata.

4. Roadmap phases, deliverables, and gates

Phases beyond Markets are conditional. Each phase ships only if security review, SOPs, evidence checklists, pilot counterparties, and risk caps are in place.

Phase	Status	Goal	Deliverables (summary)	Risk controls
Phase 1 — Markets	Committed	Credibility + access	Docs (EN/TR), verification checklist, wallet proofs, transparent LP incentives	Small-scale testing; consistent disclosures
Phase 2 — Settlement MVP	Planned (Conditional)	Shared ledger + disputed-portion vaults (pilot)	Deal Ledger; Evidence Vault; USDC allowlist; Disputed Portion Vaults (commission disputes, port costs, bunkers, laytime)	Pilot caps; conservative permissions; independent review; no auto timeouts
Phase 3 — Charter modules	Planned (Conditional)	TCT/VOY dispute templates	Deduction→Vault flows for off-hire, performance, bunkers; Laytime settlement vault; War risk/port closure extra cost vault	Evidence Standard (ESP); CAP rules; escalation playbook
Phase 4 — Claims & reporting	Planned (Conditional)	SOA support + audit trail exports	Dispute logs; reporting packs; accounting exports; optional hash anchoring	Privacy-by-design; minimal on-chain data; incident response
Phase 5 — Advanced escrow flows	Exploratory	Partial automation (only after proven pilots)	Optional: hire drip; voyage milestone auto release; more integrations	Strict gating; monitoring; rollback plan
Phase 6 — Dispute liquidity (pool)	Exploratory (Last)	Optional liquidity layer	Insurance/pool concepts considered only after adoption; separate review	Independent risk mgmt; TVL caps; avoid perverse incentives

Go / no-go gates
Safe transition principle: start with low-risk line items (port costs, bunker deltas, laytime deltas) before touching full hire/freight flows. Avoid automatic timeout releases in early phases.

5. System primitives

MRT models each fixture as one Deal Ledger with a set of Vault Line Items. Each line item has: purpose, funding rule, cap, evidence checklist, signers, and resolution path.

5.1 Deal Ledger

- Single shared page per deal: recap, CP references, key dates, payment references (SWIFT/USDC), and event timeline.
- Every action is time-stamped; settlements and counteroffers are recorded (audit trail).
- Documents remain off-chain (privacy). Optional hash anchoring can be enabled for audit purposes.

5.2 Vault Line Items

A vault line item is a narrowly scoped custody bucket. It is not a general escrow for all payments. In the MVP, hire and freight may continue off-platform; vaults are used for disputed portions.

Line item	Used for	Funding rule (MVP)	Release rule
Disputed Amount Vault	Any disputed amount (e.g., off-hire deduction, laytime delta)	Depositor funds disputed portion (instead of withholding)	Release on agreement or binding decision
Bunker Adjustment Vault	BOD/BOR/ROB delta; off-spec holdback	Party owing the disputed delta	Release based on survey/lab/decision
Port Cost Vault	DA advance + invoice approval; disputed invoice portion	Owner funds advance; disputed portion locked	Approved invoices released; disputed remainder locked
Commission Schedule (no vault)	Commission typically netted from hire	Recorded + authorized as deduction schedule	Tracking + proofs; vault only if disputed
Performance Reserve (optional)	Speed/consumption small claims	Mutual funded 50/50; capped	Pro-rata return; pays out only on decision
War Risk / Extra Cost	Premium delta; escorts; closure costs	Undisputed paid; disputed secured with CAP + optional LoU/BG	Release via agreement/decision

5.3 Evidence Standardization Protocol (ESP)

To reduce ambiguity, each dispute type uses a minimum evidence checklist. MRT does not interpret CP law; it ensures the evidence pack is complete and time-stamped.

6. Time Charter modules (TCT)

MVP principle: hire flows remain unchanged unless counterparties explicitly opt into advanced escrow flows. MRT focuses on neutralizing deductions and disputed adjustments.

6.1 Off-hire (deduction → vault)

Example: Hire 18,000/day. Charter claims 2 days off-hire = 36,000.

12. Hire is paid as normal (e.g., 15 days in advance via bank).
13. Charter raises off-hire claim and uploads evidence (notice, log extracts, noon reports).
14. At the next hire payment, Charter pays Owner net-of-claim and funds the disputed portion into Off-hire Vault:
    - Charter → Owner: 270,000 - 36,000 = 234,000
    - Charter → Off-hire Vault: 36,000
15. Negotiation window: proposals & counteroffers recorded.
16. If agreed: Owner+Charter sign to release/refund.
    If not: escalate via CP venue; on binding decision, execute via Party + Decision Executor.

Rationale
Traditional practice allows deduction with cash kept by Charter. MRT asks the deducted portion to be held neutrally instead, removing leverage and accelerating settlement.

6.2 Performance (speed/consumption) claims (optional capped reserve)

Performance claims are usually period-end calculations. Rather than holding a large percentage, MRT uses a small, capped, mutually funded reserve only when counterparties want faster closeout.

Default parameters (pilot)

- Reserve ON/OFF per deal (optional).
- Reserve target: min(Period Hire × 1%, 10,000).
- Funding: 50/50 Owner + Charter (neutral).
- Unused funds return pro-rata; payout only on agreement or binding decision.

Example scenario

Period hire: 225,000. Reserve target: 2,250 (1,125 each). Claim: 1,800.

17. Owner rejects claim → reserve remains locked.
18. Negotiation window runs; if no agreement, escalate.
19. On decision: vault pays Charter 1,800; remaining 450 returns to Owner (pro-rata outcome).

6.3 Bunkers (BOD/BOR/ROB and off-spec)

Example: Redelivery ROB dispute 20 MT @ 650 = 13,000.

20. Party owing the disputed delta funds Bunker Adjustment Vault for 13,000.
21. Survey report is uploaded; if off-spec alleged, lab results are attached.
22. Agreement or binding decision executes payout/refund (2-of-3).

6.4 Commissions (shipping practice: netting, not separate payments)

Commissions are typically netted from hire statements. The MVP starts with a Commission Schedule + Deduction Authorization rather than forcing a separate commission escrow.

- Deal records commission parties, rates, and calculation base (commission schedule).
- Owner + Charter approve the schedule (deduction authorization).
- Hire statements can be generated as Gross / Commissions / Net due.
- If commission becomes disputed, only the disputed commission amount can be placed in a Disputed Amount Vault.

6.5 War risk / port closure (Hormuz-type disruption)

War risk events create disputes over additional premiums, safe orders, waiting time, and extra costs. MRT supports: (1) pay-undisputed-now, (2) secure-disputed-with-CAP, and (3) optional documentary security (LoU/BG).

Example: Additional premium invoice 250,000. Charter accepts 120,000, disputes 130,000.

23. Undisputed payment: Charter → Owner/Underwriter: 120,000 (normal).
24. Disputed security: Charter secures part of the disputed delta via War Risk Cost Vault, capped by CAP:
    CAP = min(10% of period hire, 50,000). For a 270,000 period hire, CAP = 27,000.
    - Charter → WarRiskVault: 27,000
25. Remaining disputed security (optional): LoU / BG or agree to escalate.
26. Negotiation window is enforced; if unresolved, escalate to arbitration; execute binding decision via Party + Decision Executor.

CAP principle
CAP avoids the 'Charter locks 130k in cash' problem. Only capped cash security is held in vault unless counterparties choose otherwise.

7. Voyage Charter modules (VOY)

Voyage payments are naturally milestone-based. The MVP focuses on separating freight from laytime so one disagreement does not block everything.

7.1 Freight milestones (optional early; common 20/60/20)

Freight milestones can be supported as an optional escrow workflow once pilots prove stable. If enabled, only the unreleased remainder is frozen during a dispute.

Example: Freight 1,000,000 with 20/60/20 releases.

27. Charter funds Freight Vault with 1,000,000 (if enabled).
28. NOR/arrival → release 200,000.
29. BL issued → release 600,000.
30. Discharge complete → release remaining 200,000; if dispute arises, freeze only unreleased portion.

7.2 Laytime / Demurrage / Despatch (dedicated laytime vault)

Laytime should not be mixed with freight. Use a dedicated Laytime Settlement Vault for the disputed delta.

Example: Owner claims demurrage 80,000; Charter agrees 20,000 and disputes 60,000.

31. Charter pays undisputed portion: Charter → Owner: 20,000.
32. Charter funds Laytime Vault with disputed delta: 60,000 (neutral custody).
33. SOF/NOR/port logs are uploaded (off-chain).
34. Agreement → Owner+Charter sign to release; otherwise escalate and execute binding decision.

7.3 Laycan miss / cancellation

If voyage is cancelled (laycan missed), unreleased freight stays in vault (if funded) and is refunded. If CP provides cancellation fee/deadfreight, apply as parameterized split after agreement/decision.

7.4 Cargo damage / shortage (claims holdback)

Cargo claims can surface late. MRT supports a capped holdback (optional) so final closeout is not hostage to open-ended exposure.

Example: Closeout remainder 100,000; potential cargo exposure 60,000.

35. Release undisputed remainder: 40,000 to Owner.
36. Hold 60,000 in Cargo Claims Vault pending survey/P&I outcome.
37. On decision/settlement, release to Owner or refund/split as appropriate.

7.5 War risk / port closure (voyage)

Extra costs (escorts, deviations, port closures) are handled via separate Extra Cost Vaults with CAP and documentary security options, following the same principle as Time Charter.

8. Claims & dispute catalog (event → funding → vault behavior → evidence → resolution)

This catalog is the operational core: it tells teams exactly what to do when an event happens. It is designed to prevent funds being used as leverage.

Dispute type	Where	Default funding (MVP)	Vault behavior	Minimum evidence	Resolution path
Off-hire	TCT	Deducted amount placed into Off-hire Vault at next hire payment	Lock disputed portion; no auto release	Notices; logs; noon reports; CP refs	Negotiate → Escalate → Execute
Performance (speed/consumption)	TCT	Optional mutual reserve (capped) or disputed amount vault	Lock reserve/disputed portion	Calc summary; noon reports; baseline; weather	Negotiate → Decision → Execute
Bunkers qty/price (BOD/BOR/ROB)	TCT	Party owing delta funds Bunker Adjustment Vault	Lock delta; allow split releases	Survey; ROB; BDN; CP refs	Negotiate → Decision → Execute
Bunker off-spec/contamination	TCT/VOY	Holdback disputed portion into Off-spec Vault	Freeze portion until lab results	BDN; lab tests; survey notes	Negotiate → Decision → Execute
Laytime/demurrage delta	VOY	Undisputed paid; disputed delta into Laytime Vault	Freeze delta; keep freight separate	SOF; NOR validity; port logs	Negotiate → Decision → Execute
Cargo damage/shortage holdback	TCT/VOY	Optional capped holdback from closeout	Holdback vault; remainder released	Survey; BL remarks; P&I refs	Decision/ settlement → Execute
Port costs / DA advance	Any	Owner funds Port Cost Vault; disputed invoice portion locked	Approved invoices released; dispute locked	Invoices; receipts; approvals	Negotiate → Decision → Execute
Commissions (netting)	TCT	Schedule + authorization; vault only if disputed	If disputed, lock disputed commission amount	Fixture recap; clause; statements	Negotiate → Decision → Execute
War risk premium/extra costs	TCT/VOY	Undisputed paid; disputed secured with CAP + LoU/BG option	WarRisk/ExtraCost vault; CAP; no forced full lock	Insurer invoices; circulars; orders/response	Negotiate → Escalate → Execute

Design principle
Important: the vault does not decide who is right. It enforces custody of the disputed portion and executes agreed or binding outcomes. Arbitration remains external.

9. Technical architecture (minimum viable stack)

Start simple. The MVP must be easy to review and hard to break.

9.1 On-chain modules (MVP)

- EscrowFactory: creates vault instances with template parameters (purpose, cap, signers, timelines).
- Vault contracts: deposit, propose split, approve (2-of-3), freeze/unfreeze, execute payout/refund.
- USDC allowlist: accept only the intended USDC token address to reduce counterfeit risk.
- Events: every action emits an on-chain event for audit trail.

9.2 Off-chain modules (MVP)

- Simple dApp: Create Deal, add line items, Deposit, Propose Settlement, Approve, Export Case File.
- Evidence storage: off-chain documents; optional hash anchoring for audit trail.
- Reporting: dispute frequency, median time-to-resolution, settlement accuracy.

9.3 Security baseline

- Small pilot caps; conservative permissions; no aggressive automation/timeouts early.
- Independent review before pilot go-live.
- SOP: required evidence per dispute type + who can act as Decision Executor.
- Incident response: pause/freeze policy if stablecoin de-peg or network disruption occurs.

10. Operational rollout (pilots, limits, metrics)

Adoption in shipping is conservative. The goal is not to replace banks overnight; it is to reduce friction and improve certainty where it hurts most.

10.1 Pilot strategy

- Start with one real counterparty pair (Owner + Charterer) and low-risk line items (port costs, bunker deltas, laytime deltas).
- Keep caps small (per-deal and total).
- Measure time-to-resolution and iterate on evidence checklists (ESP).

10.2 Metrics to track

Metric	Why it matters
Disputes per transaction	Should decrease as evidence standards mature
Median time-to-resolution	Primary speed KPI vs email/Excel
Settlement accuracy	How often payouts match later decisions
User adoption	Repeat usage by brokers/ops teams
Operational incidents	Security or process failures

10.3 How to communicate (non-hype)

- We do not replace the Charter Party; we secure the settlement path for disputed portions.
- We do not replace arbitration; we accelerate negotiation and execute binding outcomes.
- We keep sensitive documents off-chain while preserving an auditable timeline.

Appendix A — Default parameters (MVP)

Parameter	Default
Owner response window	3–5 days
Negotiation window	7–14 days
CAP (war risk / extra cost vault)	min(10% of period hire, 50,000)
Performance reserve target (optional)	min(1% of period hire, 10,000), funded 50/50
Timeout behavior	No automatic release; escalate and keep funds locked

Appendix B — Evidence checklists (minimum)

Off-hire (TCT)

- Off-hire notice timestamp + date range
- Noon reports for affected period
- Relevant log extracts (engine/bridge)
- CP clause reference
- Class attendance / service report (if mechanical)

Laytime (VOY)

- Statement of Facts (SOF)
- NOR validity evidence
- Port time logs + exceptions
- Laytime terms excerpt

War risk / port closure

- Underwriter invoice(s) + coverage notice
- Circulars / advisories / port info
- Orders, refusal notices, and alternative order correspondence
- AIS/anchorage logs (if relevant)

Appendix C — Glossary

Term	Definition
Deal Ledger	Single shared deal page with timeline, payments, evidence, and settlement history.
Vault line item	Narrow-purpose custody bucket for a specific disputed amount.

Disputed portion	The delta amount under dispute; kept neutral (not held by either party).
Decision Executor	Neutral role that does not judge merits; executes agreed or binding outcomes.
ESP	Evidence Standardization Protocol: minimum evidence pack per dispute type.
CAP	Maximum cash security required in vault for certain disputes (e.g., war risk premium deltas).

Disclaimer

This document is a product roadmap intended to communicate direction and design intent. It does not constitute a binding commitment, and the scope and sequencing may change as the project learns from the market.

- Execution will be step-by-step. We prioritize proving the MVP in real market conditions (Deal Ledger, Evidence Vault, and Disputed-Portion Vaults) before expanding to more complex modules.
- All timelines are indicative only. Delivery depends on market traction, partner and compliance readiness, technical feasibility, and available resources.
- Advanced features (e.g., liquidity-style dispute finance) are optional and should be pursued only after the core workflows demonstrate clear value in live trades.

Nothing in this roadmap constitutes legal, financial, or investment advice.